

26.1.2 Biasing Circuit

When designing the bias circuits earlier, Fig. 20.47 for example, our goals were to provide biasing for general analog design. Here, in this chapter, our goal is to design a bias circuit for our op-amp designs with minimum power dissipation. When designing biasing circuits fully-differential topologies have some advantages over the single output op-amps presented in Ch. 24. For example, our earlier bias circuit provided biasing, V_{pcas} and V_{ncas} , for the floating current sources used in the class AB output stages (see Fig. 24.29 for example). These bias voltages won't be needed here.

Towards understanding this last statement, consider the two-stage op-amp seen in Fig. 26.2. Consider what happens if the noninverting op-amp input, v_p , increases relative to the inverting input, v_m . The drain voltage of M1R drops while the drain voltage of M1L rises. The decrease in the gate voltage of M6R causes it to turn on and the output to go high. At the same time, the increase in the drain voltage of M1L causes M4R to shut off and thus so does M7R (giving class AB operation). In simple terms, we can yank the gates of M4 or M6 down independent of the diff-amp's tail current.

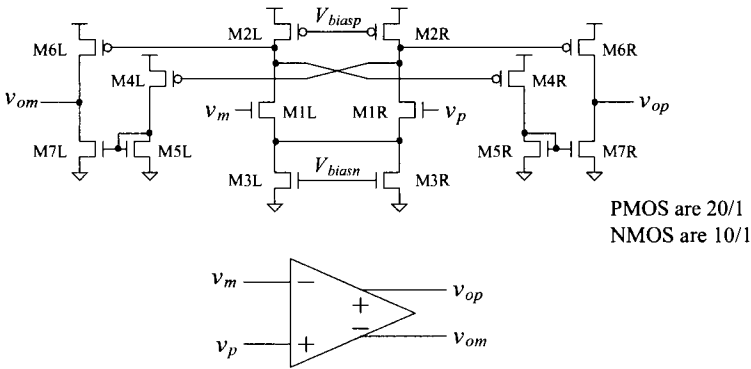


Figure 26.2 A two-stage fully-differential op-amp. Compensation and CMFB are not shown. Output stage operates class AB. See discussion in the next section concerning the output voltage of the diff-amp.

Layout of Differential Op-Amps

One of the common things we do in this chapter is draw schematics that are symmetrical. If we were to draw a line down the middle of the op-amp in Fig. 26.2, separating the left and right sides of the schematic, we could fold the left side of the schematic directly over onto the right side of the schematic and see a perfect match. For example, instead of drawing the diff-amp's tail current source (M3) as a single MOSFET with twice the width, we've drawn it as two MOSFETs in parallel, each having the same width. *Drawing schematics in this fashion is useful when doing layout.* We can fold the schematic in half and lay out like devices, e.g., M6L and M6R, directly next to each other. Further, then the outputs (and inputs) of the op-amp are laid out right next to each other.

Self-Biased Reference

Figure 26.3 shows the bias circuit we'll use in this chapter (see also Fig. 20.22 and the associated discussion). We used the length of 2 MOSFETs for the added amplifier to minimize power dissipation and boost the amplifier's gain. The current pulled from VDD is approximately $50\text{ }\mu\text{A}$. Figure 26.4 shows the simulation results of how the current varies with changes in VDD . Notice how V_{biasn} is close to the 400 mV , and V_{biasp} is close to $VDD - 400\text{ mV}$ (of course the absolute value of V_{biasp} varies with VDD but the ideal value of V_{SG} for the PMOS devices will be 400 mV). Note that, as discussed in Sec. 20.1.4, it is important to ensure that the reference is stable over all possible operating conditions and loads (connected to V_{biasn} and V_{biasp}). Notice that we aren't discussing how the reference currents vary with process shifts in the MOSFETs and the resistor. This is an important practical concern.

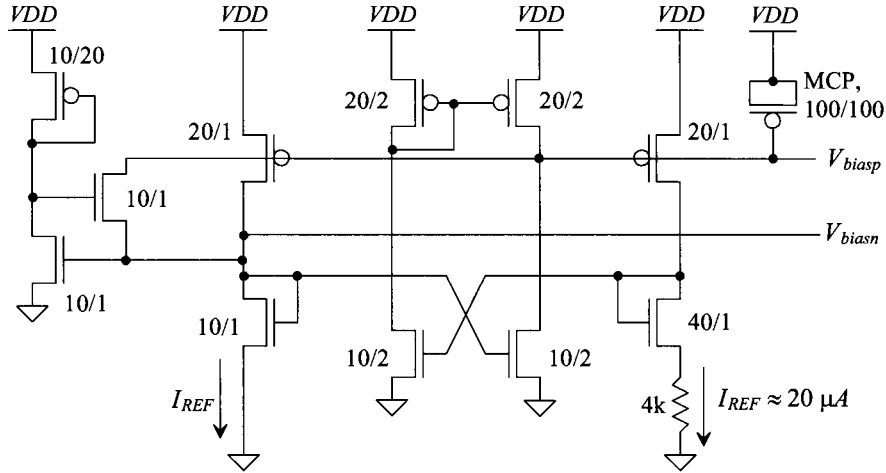


Figure 26.3 Biasing circuit used in this chapter. This bias circuit pulls approximately $50\text{ }\mu\text{A}$.

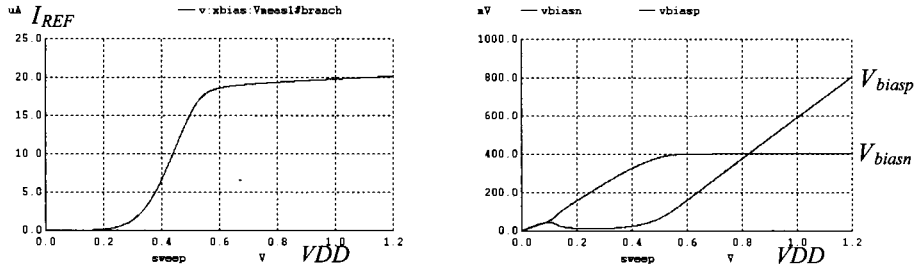


Figure 26.4 Simulating how the reference current changes with VDD .

26.2 Basic Concepts

Before we discuss the design of op-amps, let's look at some basic concepts that will be useful when evaluating a specific op-amp topology.

Modeling Offset

In a real circuit, especially an analog circuit designed with minimum length devices, mismatches can be a significant factor in the selection of an op-amp topology. In a SPICE simulation, all of the MOSFETs are perfectly matched. We have to come up with a method, in SPICE, to determine an op-amp's sensitivity to offsets.

Consider the circuit in Fig. 26.5. In (a) the measured currents should be equal. Both MOSFETs V_{SG} and V_{SD} are equal (and they are the same size). However, for whatever reason (e.g., threshold voltage mismatch), there exists a mismatch between the two devices that causes a difference in their drain currents. We can model this mismatch in a SPICE simulation, as seen in (b), by adding a DC voltage source in series with the gate of M2. For a general design, we can insert these offset voltages at various points in the circuit and verify that the circuit still functions correctly. The next question that needs answering is: "What value of V_{os} should be used?" While no absolute answer can be given here, **we'll use a V_{os} of 50 mV** (5% of V_{DD} or roughly 20% of the threshold voltage). The reader might feel that this value is way too high. However, if we can design circuits that function properly over the process, voltage, and temperature (PVT) variations with an offset this large, it is likely the op-amp will be difficult to "break."

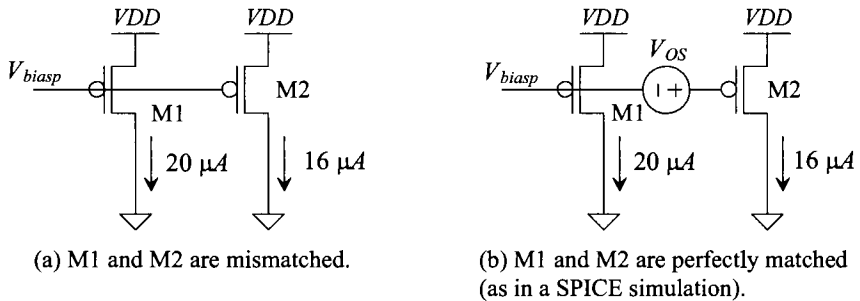


Figure 26.5 How we add an offset into the circuit to model mismatch.

A Diff-Amp

Figure 26.6a shows a basic diff-amp without an offset. This offset-free diff-amp has simulated output voltages of 700 mV. The diff-amp in (b) is simulated with an input-referred offset voltage of 50 mV. The polarity of the offset doesn't matter because it simply swaps the two output voltage values. In other words, we could have put the same polarity offset on the other input of the diff-amp and swapped the voltages on the output of the op-amp. We put the offset voltage on the op-amp's input because the input has the greatest effect on the diff-amp's output voltages. Note: we can think of the 50 mV as an input signal causing an output signal difference, from the ideal 700 mV, of 900 mV – 550 mV or 350 mV (assuming linear operation). This indicates the gain of the diff-amp is only 7!

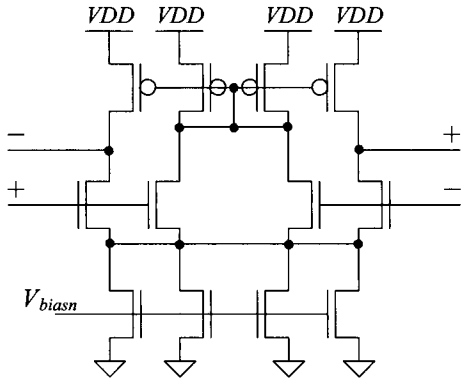


Figure 26.7 A fully-differential diff-amp that generates its own bias for the PMOS.

fully-differential op-amp topology (both the inputs and the outputs of the op-amp, with feedback, are double-ended signals swinging around V_{CM}), the input common-mode voltage of the op-amp is constant ($= V_{CM}$). The common-mode rejection ratio isn't as important when the input common-mode voltage of the op-amp doesn't vary.

Using a CMFB Amplifier

Another possible way to set the diff-amp's output voltages to a known value is seen in Fig. 26.8. An amplifier (called a common-mode feedback amplifier or CMFB amplifier) is used to amplify the difference between the average of the diff-amp's outputs and V_{biasp} . If the gain of the CMFB amplifier is large, then the average of the two outputs will be very close to V_{biasp} . Note that the CMFB amplifier's output signal, V_{CMFB} , is common, through M1L and M1R, to both outputs. Any variation in V_{CMFB} affects each output by the same amount. This is important because all we want the CMFB amplifier to do is make

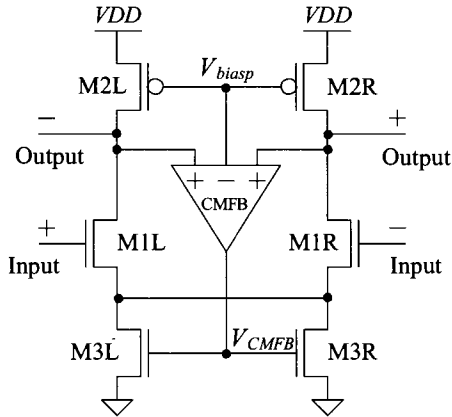
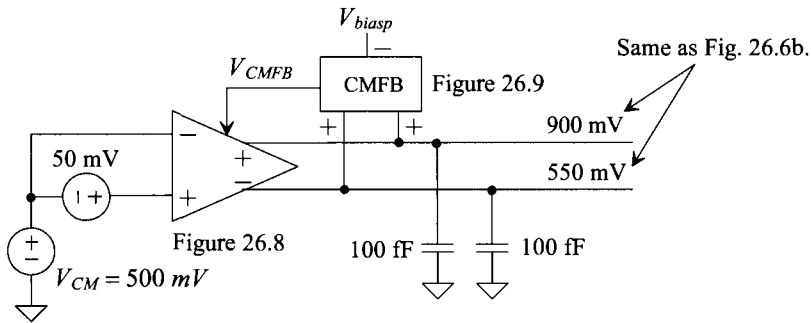
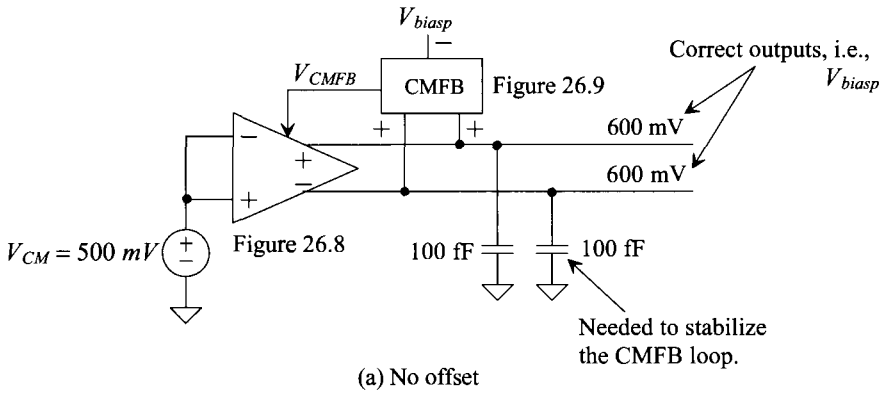


Figure 26.8 Using a common-mode feedback (CMFB) amplifier to set the output voltages.



(b) With a 50 mV offset. Note how the CMFB isn't doing anything.

Figure 26.10 Simulating the operation of the CMFB circuit in Fig. 26.9.

Compensating the CMFB Loop

Consider the schematic seen in Fig. 26.11. The diff-amp behaves like an operational transconductance that can be compensated, as discussed in Sec. 24.3 (see Eq. [24.44]). The CMFB loop can be compensated in a similar fashion. The AC common-mode signal is represented in this schematic as v_c . If the gain of the CMFB amplifier is A_{cm} , then following the procedure leading to Eq. (24.44), we can write the unity-gain frequency of the CMFB loop as

$$f_{un,cm} = \frac{A_{cm} \cdot g_{mn}}{2\pi C_L} \quad (26.2)$$

If we want to compensate the CMFB loop with the same load capacitance used to compensate the differential forward signal path, then we must ensure that the gain of the CMFB amplifier is less than or equal to unity,

$$A_{cm} \leq 1 \quad (26.3)$$